

MODELO ARX PRUEBA ESCALÓN

IDENTIFICACION DE PROCESOS USANDO MINIMOS CUADRADOS

```
clear all
close all

Datos=xlsread('Datos_Escalon - Copy.xlsx');

t=Datos(:,1);
U=Datos(:,2);
Y=Datos(:,4);

subplot(2,1,1)
plot(t,U,'* k','MarkerSize',4)
ylabel('Entrada')
grid on
subplot(2,1,2)
plot(t,Y,'o b','MarkerSize',4)
ylabel('Salida')
xlabel('Tiempo')
hold on
grid on

% Seleccionando na=5, nb=5, d=1
s=length(Y);

X=[-[0;Y(1:s-1)],-[0;0;Y(1:s-2)],-[0;0;0;Y(1:s-3)], [0;U(1:s-1)], [0;0;U(1:s-2)], [0;0;0;U(1:s-3)],...
% X=[-[0;Y(1:s-1)],-[0;0;Y(1:s-2)],-[0;0;0;Y(1:s-3)],...
%    -[0;0;0;0;Y(1:s-4)],-[0;0;0;0;0;Y(1:s-5)],...
%    [0;U(1:s-1)], [0;0;U(1:s-2)], [0;0;0;U(1:s-3)],...
%    [0;0;0;0;U(1:s-4)], [0;0;0;0;0;U(1:s-5)], [0;0;0;0;0;0;U(1:s-6)]];

%Matriz de covarianzas
Q=X'*X
```

```
Q = 7x7
10^4 x
    2.9678    2.8089    2.6401   -0.0738   -0.0738   -0.0719   -0.0688
    2.8089    2.8257    2.6194   -0.0701   -0.0701   -0.0701   -0.0682
    2.6401    2.6194    2.5730   -0.0650   -0.0650   -0.0650   -0.0650
   -0.0738   -0.0701   -0.0650    0.0020    0.0019    0.0018    0.0017
   -0.0738   -0.0701   -0.0650    0.0019    0.0019    0.0018    0.0017
   -0.0719   -0.0701   -0.0650    0.0018    0.0018    0.0018    0.0017
   -0.0688   -0.0682   -0.0650    0.0017    0.0017    0.0017    0.0017
```

```
%Vector de parámetros
Theta=inv(Q)*X'*Y
```

```
Theta = 7x1
    0.3205
```

```

0.3732
-0.1128
18.8496
18.6068
14.2038
12.0637

```

```

z=tf('z',0.1);
Gpm=((Theta(4)*z^3+Theta(5)*z^2+Theta(6)*z+Theta(7))/(z^3+Theta(1)*z^2+Theta(2)*z+Theta(3)))*(z^5+Theta(1)*z^4+Theta(2)*z^3+Theta(3)*z^2+Theta(4)*z+Theta(5))

```

Gpm =

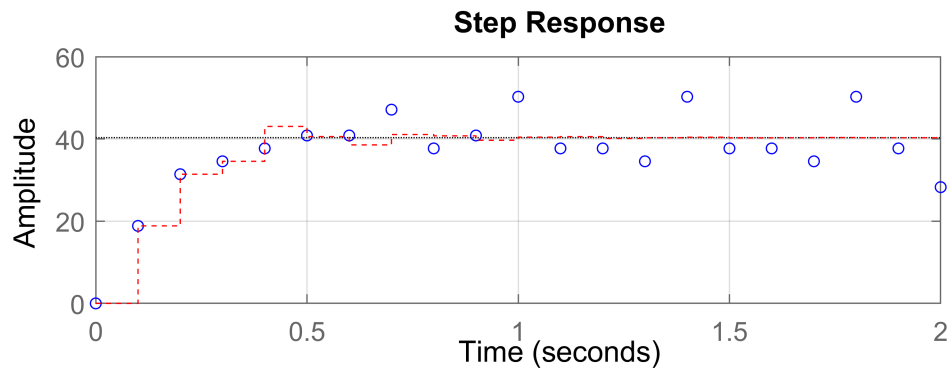
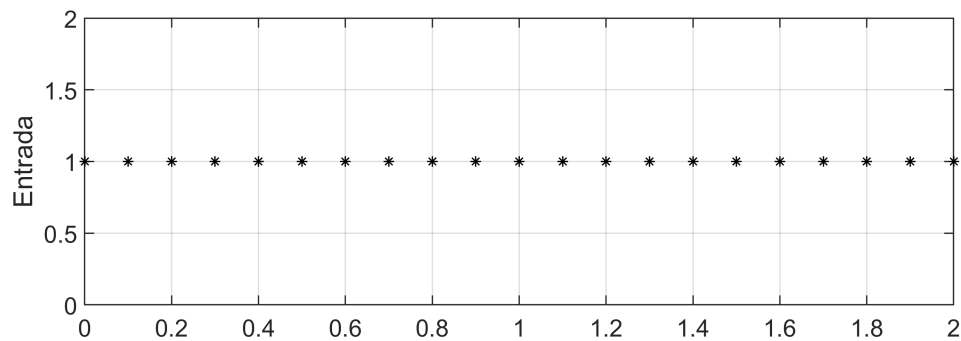
$$\frac{18.85 z^3 + 18.61 z^2 + 14.2 z + 12.06}{z^4 + 0.3205 z^3 + 0.3732 z^2 - 0.1128 z}$$

Sample time: 0.1 seconds
Discrete-time transfer function.

```

%Gpm=((Theta(6)*z^5+Theta(7)*z^4+Theta(8)*z^3+...
%   Theta(9)*z^2+Theta(10)*z+Theta(11)) / ...
%   (z^5+Theta(1)*z^4+Theta(2)*z^3+Theta(3)*z^2+...
%   Theta(4)*z+Theta(5)))...
%   *(z^(-1))
step(Gpm,'r--',2)
grid on

```



```

%Ym=step(Gpm,14);
Ym = lsim(Gpm,U,t)

```

```
Ym = 21x1
    0
  18.8496
  31.4159
  34.5575
  43.0500
  40.5732
  38.5515
  41.0815
  40.7460
  39.6812
    ⋮
```

```
%Validación del modelo
Sy=sum((Y-mean(Y)).^2)
```

```
Sy = 2.4965e+03
```

```
Sr=sum((Y-Ym).^2)
```

```
Sr = 619.6076
```

```
R2=(Sy-Sr)/Sy
```

```
R2 = 0.7518
```